Lesson notes · for you

Wrong examples, wrong answers

AI security · Primary school · Lesson 2

Time
20 min

Materials
Ten small cards with simple drawings of two kinds of fruit (or real leaves of two kinds); A pencil to write a label on each card

Learning goals

- Explain how AI learns from examples with labels.
- Show what wrong labels do to an AI.
- Find and fix the odd ones out.

Activity: Teach the sorter

1. Make (5 min): Draw ten small cards: five of one fruit and five of another, such as mango and lemon. Write the right label on each card.
2. Teach (5 min): Your friend is the “computer”. Show them all ten cards with their labels. Then hold up five new drawings and ask them to sort each one.
3. Trick (5 min): Make one new card with the wrong label, for example a big mango labelled “lemon”. Teach your friend again with this card mixed in, then sort new drawings again. Does your friend make more mistakes?
4. Find (5 min): Lay all the cards out together. Which ones look odd next to the others with the same label? Fix those labels, and talk about how you found them.

No cards? Use real leaves of two kinds, or stones of two colours.

Discussion

- Why can't a computer tell when a label is wrong?
- Why might someone add wrong examples on purpose?
- How can you help keep examples right?

Check yourself

- Explain in one sentence why wrong labels make wrong answers.

Try it this week
Play “Teach the sorter” at school or at home: one friend is the computer and learns from picture cards with labels. Sneak in one wrong label, and see what the “computer” gets wrong.

Worksheet

Wrong examples, wrong answers

AI security · Primary school · Lesson 2

Name _____ Class _____ Date _____

1 **Your class is teaching a computer to sort leaves. A friend labels a few wrongly, as a joke. What should you do?**

- ◯ Nothing: it's only a few leaves
- ◯ Add more wrong labels for fun
- ◯ Fix the labels and tell your teacher

2 **Put in order: how a computer learns to sort fruit.**

- ☐ Give each picture a label
- ☐ It sorts new fruit by itself
- ☐ Collect pictures of fruit
- ☐ The computer finds what's alike

3 **A computer can always tell when one of its labels is wrong.**

- ◯ True
- ◯ False

4 **Who could put wrong labels into the examples? Choose all that apply.**

- ☐ A friend labelling for a joke
- ☐ Someone playing a trick on purpose
- ☐ Someone in a hurry
- ☐ A careful helper who checks twice

5 **Match each problem to its fix.**

A label written in a hurry	• • • •	Ask where they came from
A strange example among the others	• • • •	Check it twice
Examples from someone you don't know	• • • •	Look closer, then fix or remove it

✎ **Have you ever learned something wrong because someone told you wrong? How did you find out?**

Check your answers

Wrong examples, wrong answers

AI security · Primary school · Lesson 2

1 **Fix the labels and tell your teacher**

Even a few wrong labels teach the computer mistakes.

2 **Top to bottom: 2, 4, 1, 3**

First the pictures, then a label on each.

3 **False**

The labels are its teacher. It can't tell a wrong label from a right one, so it learns the mistake.

4 **Someone in a hurry; Someone playing a trick on purpose; A friend labelling for a joke**

Wrong labels come from hurry, from jokes, and sometimes from someone spoiling the AI on purpose.

5 **A label written in a hurry — Check it twice**
A strange example among the others — Look closer, then fix or remove it
Examples from someone you don't know — Ask where they came from

Careful checking, a close look at odd ones out, and knowing where examples came from all keep wrong labels out.